

with vectors as in Examples 1 and 2, we say that the solution is in parametric vector form.

Solutions of Nonhomogeneous Systems

When a nonhomogeneous linear system has many solutions, the general solution can be written in parametric vector form as one vector plus an arbitrary linear combination of vectors that satisfy the corresponding homogeneous system.

Example 3: Describe all solutions of $Ax=b$, where

$$A = \begin{bmatrix} 3 & 5 & -4 \\ -3 & -2 & 4 \\ 6 & 1 & -8 \end{bmatrix} \text{ and } b = \begin{bmatrix} 7 \\ -1 \\ -4 \end{bmatrix}$$

Solution: By Theorem 3, the solution of $Ax=b$ is the same as the solution of the system of equations corresponding to the augmented matrix $[A \ b]$.

$$\begin{array}{l} \text{Row 2} = \text{Row 2} + \text{Row 1} \\ \text{Row 3} = \text{Row 3} - 2 \times \text{Row 1} \\ \text{Row 2} = \text{Row 2} \times \frac{1}{3} \\ \text{Row 1} = \text{Row 1} \times \frac{1}{3} \end{array} \begin{array}{l} \begin{bmatrix} 3 & 5 & -4 & 7 \\ -3 & -2 & 4 & -1 \\ 6 & 1 & -8 & -4 \end{bmatrix} \xrightarrow{\text{Row 2} = \text{Row 2} + \text{Row 1}} \begin{bmatrix} 3 & 5 & -4 & 7 \\ 0 & 3 & 0 & 6 \\ 6 & 1 & -8 & -4 \end{bmatrix} \\ \xrightarrow{\text{Row 3} = \text{Row 3} - 2 \times \text{Row 1}} \begin{bmatrix} 3 & 5 & -4 & 7 \\ 0 & 3 & 0 & 6 \\ 0 & -9 & 0 & -18 \end{bmatrix} \\ \xrightarrow{\text{Row 2} = \text{Row 2} \times \frac{1}{3}} \begin{bmatrix} 3 & 5 & -4 & 7 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ \xrightarrow{\text{Row 1} = \text{Row 1} - 5 \times \text{Row 2}} \begin{bmatrix} 3 & 0 & -4 & -3 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \\ \xrightarrow{\text{Row 1} = \text{Row 1} \times \frac{1}{3}} \begin{bmatrix} 1 & 0 & -4/3 & -1 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array}$$

Same as that of Example 1, it's easy to see that the final matrix is in ref form. Column 3 has no pivots. $\therefore$ The variable x_3 is free. The corresponding system of equations are: $x_1 - \frac{4}{3}x_3 = -1$ and $x_2 = 2$. $\therefore$ The solution set is: $x_1 = \frac{4}{3}x_3 - 1$, $x_2 = 2$ and x_3 is a free variable.